Writing theoretical and statistical hypotheses

Each table has two rows (two-sided and one-sided) and two hypothesis columns (theoretical and statistical). H_0 is the null hypothesis; H_1 is the alternative.

1 One-sample t-test

Tail/side	Theoretical hypothesis	Statistical hypotheses
Two-sided	Amsterdam residents spend, on average, a different amount of time using media than five hours per day.	$H_0 : \mu = 5$; $H_1 : \mu \neq 5$
One-sided	Amsterdam residents spend, on average, more than five hours per day using media.	$H_0 : \mu \leq 5$; $H_1 : \mu > 5$

Symbol: μ is the population mean daily media-use time of Amsterdam residents, in hours.

2 Paired-sample t-test

Measure both regional- and national-newspaper reading time for each resident.

Tail/side	Theoretical hypothesis	Statistical hypotheses
Two-sided	Dutch residents spend, on average, different amounts of time reading regional and national newspapers.	$H_0 : \mu_D = 0$; $H_1 : \mu_D \neq 0$
One-sided	Dutch residents spend, on average, more time reading regional than national newspapers.	$H_0 : \mu_D \leq 0$; $H_1 : \mu_D > 0$

Symbols: D is each person's weekly reading time for ~~regional minus national newspapers~~, in minutes; μ_D is the population mean of these within-person differences.



3 Independent-sample t-test

Tail/side	Theoretical hypothesis	Statistical hypotheses
Two-sided	Populists and non-populists differ in their average belief in conspiracy theories.	$H_0 : \mu_P = \mu_N$; $H_1 : \mu_P \neq \mu_N$
One-sided	Populists have stronger average belief in conspiracy theories than non-populists.	$H_0 : \mu_P \leq \mu_N$; $H_1 : \mu_P > \mu_N$

Symbols: μ_P and μ_N are the population mean conspiracy-belief scores for populists and non-populists. Higher scores indicate stronger belief.

4 One-way ANOVA

Compare conspiracy-belief scores across three groups: people who never, sometimes, or often use social media for news.

Tail/side	Theoretical hypothesis	Statistical hypotheses
not applicable	Average belief in conspiracy theories differs among the three social-media news-use groups.	$H_0 : (\mu_1 = \mu_2) \wedge (\mu_1 = \mu_3) \wedge (\mu_2 = \mu_3)$; $H_1 : (\mu_1 \neq \mu_2) \vee (\mu_1 \neq \mu_3) \vee (\mu_2 \neq \mu_3)$

Symbols: μ_1, μ_2, and μ_3 are the population mean belief scores for the never, sometimes, and often groups. $\wedge$ means "and"; $\vee$ means "or" (at least one statement is true).

The ANOVA statistical hypothesis can be written in words (required in the exam):

H0: three population means are equal;

H1: at least one pair of population means differs

Note: ANOVA testing group mean differences reports one-sided p value (more precisely, right-sided p-value). However, this right-sided p-value does not refer to the direction of the differences. So it is confusing to call it 'one-sided' hypothesis.

It also does not make sense to call it 'two-sided' because p-value is just on one side. So here I put 'not-applicable' for column `Tail\side`.

5 Two-by-two ANOVA

Two factors: avatar attractiveness (attractive/unattractive) and embodiment (high: head-mounted-display VR; low: desktop). The outcome is social participation.

5.1 Main effect of avatar attractiveness

Tail/side	Theoretical hypothesis	Statistical hypotheses
not-applicable	Embodying an attractive versus an unattractive non-humanlike avatar changes average social participation.	$H_0 : \mu_A = \mu_U$; $H_1 : \mu_A \neq \mu_U$
not-applicable	Embodying an attractive, compared to an unattractive non-humanlike avatar, has a higher average social participation.	$H_0 : \mu_A \leq \mu_U$; $H_1 : \mu_A > \mu_U$

Symbols: μ_A and μ_U are the population mean social-participation scores for attractive and unattractive avatars, each averaged equally over the two embodiment levels.

Note: Again, for ANOVA, better not to call either 'one-sided' or 'two-sided' because it does not align with the p-value.

Two-by-two ANOVA is a special case, where for each one-degree-of-freedom main effect, the p-value equals the corresponding two-sided independent sample t-test p-value.

To know the direction of the main effect, you need to compare the mean differences of the two conditions accordingly. ANOVA does not explicitly tell you that.

5.2 Interaction between attractiveness and embodiment

Tail/side	Theoretical hypothesis	Statistical hypotheses
not-applicable	The effect of avatar attractiveness on average social participation differs between high and low embodiment.	$H_0 : \Delta = 0$; $H_1 : \Delta \neq 0$
~~not-applicable~~	~~The attractive-versus-unattractive difference in average social participation is more positive under high than low embodiment.~~	~~$H_0 : \Delta \leq 0$; $H_1 : \Delta > 0$~~

Symbols: $\Delta = (\mu_{AH} - \mu_{UH}) - (\mu_{AL} - \mu_{UL})$ is the population difference of differences. The four means refer to attractive/high, unattractive/high, attractive/low, and unattractive/low conditions, respectively.

Note: The ANOVA interaction F test itself tests $\Delta = 0$ against $\Delta \neq 0$. Its p-value can be used conservatively for $H_0 : \Delta \leq 0$ against $H_1 : \Delta > 0$. To know the direction of the interaction effect, you need to compare the mean differences of the four conditions accordingly. ANOVA does not explicitly tell you that.

6 Two-variable correlation

Tail/side	Theoretical hypothesis	Statistical hypotheses
Two-sided	Daily social-media use is correlated with belief in conspiracy theories.	$H_0 : \rho = 0$; $H_1 : \rho \neq 0$
One-sided	Daily social-media use is positively correlated with belief in conspiracy theories.	$H_0 : \rho \leq 0$; $H_1 : \rho > 0$

Symbol: ρ is the population Pearson correlation between daily social-media use in hours and conspiracy-belief scores.

7 Simple linear regression

Use daily social-media use in hours to predict conspiracy-belief scores.

Tail/side	Theoretical hypothesis	Statistical hypotheses
Two-sided	Daily social-media use predicts belief in conspiracy theories.	$H_0 : \beta_1 = 0$; $H_1 : \beta_1 \neq 0$
One-sided	Daily social-media use positively predicts belief in conspiracy theories.	$H_0 : \beta_1 \leq 0$; $H_1 : \beta_1 > 0$

Symbol: β_1 is the population slope: one additional hour of daily social-media use predicts β_1 unit change in the predicted mean belief score.

8 Multiple linear regression

Use daily social-media use, age, and education to predict conspiracy-belief scores.

Tail/side	Theoretical hypothesis	Statistical hypotheses
Two-sided	Daily social-media use predicts conspiracy belief, controlling for age and education.	$H_0 : \beta_1 = 0$; $H_1 : \beta_1 \neq 0$
One-sided	Daily social-media use positively predicts conspiracy belief, controlling for age and education.	$H_0 : \beta_1 \leq 0$; $H_1 : \beta_1 > 0$

Symbol: β_1 is the population slope for daily social-media use: one additional hour of daily social-media use predicts β_1 unit change in the predicted mean belief score, while holding age and education constant.

9 ANOVA in the regression

ANOVA in the regression, imprecisely speaking, compares the improvement of linear prediction (by adding predictors) with the remaining prediction error (residuals).

9.1 Simple regression: overall model test

Model comparison: Intercept plus daily social-media model vs. Intercept only model. The intercept-only model just use sample mean of the conspiracy-belief score for everyone.

Tail/side	Theoretical hypothesis	Statistical hypotheses
not applicable	Intercept plus daily social-media model improves the prediction of conspiracy belief, compared with the intercept-only model.	$H_0 : \beta_1 = 0$; $H_1 : \beta_1 \neq 0$

Symbol: Here, β_1 is the population slope for social-media use. This ANOVA tests ends up the same conclusion from the simple regression coefficient test using t-distribution. The relation is $F = t^2$.

9.2 Multiple regression: overall model test

Model comparison: Intercept plus daily social-media use, age, and education model vs. Intercept only model.

Tail/side	Theoretical hypothesis	Statistical hypotheses
not applicable	Intercept plus social-media use, age, and education model improves the prediction of conspiracy belief, compared with the intercept-only model.	$H_0 : (\beta_1 = 0) \wedge (\beta_2 = 0) \wedge (\beta_3 = 0)$; $H_1 : (\beta_1 \neq 0) \vee (\beta_2 \neq 0) \vee (\beta_3 \neq 0)$

Symbols: β_1, β_2, and β_3 are the population partial slopes for social-media use, age, and education in the multiple-regression model. H0 sets all three slopes to zero; H1 sets at least one to be nonzero. A significant overall test does not establish which predictor contributes.

9.3 Multiple regression: added contribution of social-media use

Model comparison: Intercept plus age, education, and daily social-media use model vs. Intercept plus age and education model.

Tail/side	Theoretical hypothesis	Statistical hypotheses
not applicable	Social-media use improves the prediction of conspiracy belief beyond age and education.	$H_0 : \beta_1 = 0$; $H_1 : \beta_1 \neq 0$

Symbol: Here, β_1 is the population slope for social-media use **holding age and education constant**, which is different than β_1 from the simple regression above.

SPSS: The ANOVA table gives the overall model test. To test the added contribution of social media use, enter age and education in Block 1 and add social-media use in Block 2; request the R^2 change and its F change test.

10 Chi-square test of association in a crosstab

Netflix- and television-watching frequency each have four categories: never, sometimes, often, and very often.

Tail/side	Theoretical hypothesis	Statistical hypotheses
not applicable	Netflix-watching frequency is associated with television-watching frequency.	$H_0 : \bigwedge_{i,j=1}^{4}(p_{ij} = p_{i\cdot}p_{\cdot j})$; $H_1 : \bigvee_{i,j=1}^{4}(p_{ij} \neq p_{i\cdot}p_{\cdot j})$

Symbols: i indexes Netflix categories and j television categories, each from 1 (never) to 4 (very often). p_{ij} is the population probability of being in both categories; $p_{i\cdot}$ and $p_{\cdot j}$ are the respective row and column marginal probabilities. $\bigwedge$ means "and" across all 16 cells; $\bigvee$ means "or" across those cells. Under H_0, every joint probability equals the product of its marginal probabilities (therefore, two variables are independent). Under H_1, at least one equality fails (two variables are associated).

The Chi-square statistical hypothesis can be written in words (not required in the exam):

H0: For **every combination** of a Netflix-watching category and a television-watching category, the population probability of belonging to both categories equals the probability of belonging to the Netflix category multiplied by the probability of belonging to the television category.

H1: For **at least one combination** of a Netflix-watching category and a television-watching category, the population probability of belonging to both categories does not equal the product of those two marginal probabilities.

Note: "Every combination" corresponds to $\bigwedge$ (and across all 16 cells); "at least one combination" corresponds to $\bigvee$ (or across the cells). These are the independence and association hypotheses, respectively. The chi-square test is nondirectional.